



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	09/07/2026
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	09/07/2027

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	Haptic device - Lambda.07 - Force Dimension
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Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

This risk assessment covers routine setup, connection, calibration, operation, testing, demonstration, software development, and shutdown of the Force Dimension lambda.7 haptic device in the Kinesis CTP laboratory. The device is a powered force-feedback haptic interface with translational arms, a rotational wrist, an end-effector force gripper handle, a stand, USB computer connection, and external power supply. Activities may include assembling or repositioning the stand, connecting the original power supply and USB cable, installing or running Force Dimension SDK software, automatic calibration, use of Haptic Desk diagnostic software, and running haptic demonstration or research applications.

Reason for Risk Assessment

To identify, assess, and control risks associated with powered motion, force feedback, pinch points, electrical supply, software-driven calibration, ergonomic use, unstable setup, cables, manual handling, and equipment damage during use of the lambda.7 haptic device in the Kinesis CTP lab.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2	 Employees
2	 Students
2	 Visitors

Current Rating	Hazard Information	Task Status
Moderate Risk 	Moving Parts The lambda.7 haptic device contains powered translational arms, a rotational wrist, electromagnetic brakes, active gravity compensation, force feedback, and an end-effector handle. During calibration, diagnostic tests, demonstrations, or user-developed applications, the device may move unexpectedly or apply forces to the user. Contact with moving arms, wrist components, or the handle may cause impact, trapping, pinching, strain, or equipment collision. The manufacturer states that the automatic calibration procedure drives the device throughout its workspace and the device must not be touched during calibration.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Only trained or authorised users may operate the device. - Users must read and follow the manufacturer manual before use. - Workspace around the device must be kept clear of objects before powering or calibration. - Users must not touch the device during automatic calibration. - Forces are disabled by default when an application opens a device connection. - Electromagnetic damping/brakes and communication watchdog safety features are present. - Operators must remain ready to disable forces, stop the software, or power off the device if unexpected movement occurs. - Software applications must be tested at low force/speed settings where practicable before normal use. - The device must be switched off when not in use.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Electricity The lambda.7 uses an external power supply, universal 100-240 V mains input, power connector, USB connection, and internal electronics. Damaged cables, incorrect power supply, overloaded outlets, exposure to liquids, or unauthorised disassembly could cause electric shock, short circuit, fire, equipment damage, or unexpected device behaviour.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Device must only be operated using the original manufacturer power supply. - Non-original or damaged power supplies must not be used. - Device must not be used or placed near water. - Wall outlets and extension cords must not be overloaded. - Power supply, power cable, USB cable, connectors, and device casing are visually inspected before use. - Damaged cables, loose connectors, exposed conductors, or abnormal behaviour require the device to be removed from service. - Users must not disassemble the device or remove panels. - Repairs are carried out only by authorised CTP specialists or qualified service personnel. - Device is switched off when not in use.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	Ergonomics Use of the haptic handle, wrist, gripper, computer, and display for extended sessions may cause fatigue, awkward posture, wrist strain, shoulder strain, repetitive movement discomfort, or reduced user control. The device includes gravity compensation to reduce fatigue, but poor workstation setup or prolonged use may still create ergonomic risk.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Device height and workstation position are adjusted before use. - Users take breaks during prolonged operation or software testing. - Gravity compensation is available and enabled by default unless deliberately disabled through the SDK. - Users stop operation if discomfort, fatigue, or loss of control occurs. - Workstation setup follows general lab ergonomic guidance.	

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	Fire Fire may result from overloaded wall outlets or extension cords, damaged power supply, damaged cables, short circuit, overheating, incorrect replacement power supply, or leaving the device powered unnecessarily. The manufacturer specifically warns against overloading outlets and extension cords.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Only the original manufacturer power supply is used. - Power cables and connectors are inspected before use. - Device is switched off when not in use. - Extension cords and power strips are not overloaded. - Fire detection, emergency procedures, and fire extinguishers are available in the laboratory. - Faulty or overheating equipment is removed from service and reported.	

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.